

MIDELT, Morocco

WMO: 601950

Lat: 32.683N

Lon: 4.733W

Elev: 4970

StdP: 12.24

Time Zone: 1.00 (E01)

Period: 90-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.5	30.7	11.8	12.0	47.4	15.7	14.6	46.4	28.6	50.3	23.3	42.5	5.2	260	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	93.9	63.1	91.8	62.4	89.6	61.8	67.2	87.6	65.6	85.6	64.2	83.7	7.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	95.1	74.3	58.7	88.9	73.4	57.3	84.2	72.8	34.8	88.1	33.3	85.7	32.1	83.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.7	20.5	16.8	DB	24.3	98.0	3.5	2.3	21.8	99.6	19.7	101.0	17.7	102.3	15.2	104.0
			WB	23.3	69.9	3.5	3.4	20.7	72.4	18.7	74.4	16.7	76.3	14.2	78.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	43.6	46.7	51.5	55.5	62.1	70.8	77.7	76.6	68.8	60.5	51.4	46.0
	DBStd	12.89	5.68	6.01	6.65	6.09	7.40	6.14	4.47	4.02	5.56	5.32	6.21	5.17
	HDD50	623	210	121	63	21	4	0	0	0	0	5	58	141
	HDD65	3219	664	511	420	287	144	20	0	0	26	152	407	588
	CDD50	4034	11	30	108	186	380	625	858	824	563	330	101	18
	CDD65	1156	0	0	0	2	55	195	393	359	139	13	0	0
	CDH74	11624	0	0	12	54	559	1939	4290	3553	1128	89	0	0
	CDH80	4904	0	0	0	2	139	762	2071	1593	335	2	0	0
Wind	WSAvg	5.4	5.1	5.4	6.7	7.1	5.8	5.2	4.6	4.4	4.7	4.9	5.5	5.1
Precipitation	PrecAvg	7.5	0.5	0.6	0.7	0.6	0.8	0.5	0.3	0.5	0.8	0.9	0.8	0.6
	PrecMax	12.9	1.6	2.5	1.4	1.7	2.9	2.0	1.3	1.8	2.4	4.5	4.8	1.8
	PrecMin	4.8	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.5	0.6	0.3	0.5	0.8	0.5	0.3	0.5	0.6	1.0	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	67.7	75.3	78.9	87.7	93.8	97.2	95.3	91.0	79.8	71.7	64.5
		MCWB	47.2	48.1	51.8	55.0	59.7	61.0	64.1	64.0	62.2	57.8	53.2	46.3
	2%	DB	59.9	64.6	70.8	75.3	84.1	90.7	94.7	92.9	86.7	76.8	68.6	60.7
		MCWB	45.2	46.9	50.3	53.5	58.8	59.9	63.4	63.3	61.0	57.2	51.9	46.3
	5%	DB	56.5	61.8	67.7	71.7	80.4	87.7	92.7	90.8	84.1	74.0	65.9	57.7
		MCWB	43.5	45.4	49.2	52.0	57.9	59.9	62.9	62.7	60.0	56.3	50.8	44.5
	10%	DB	53.3	58.4	64.3	68.3	76.6	84.6	90.6	88.6	80.8	71.3	62.4	54.9
		MCWB	41.9	44.4	47.8	50.8	56.4	59.5	62.4	62.3	59.5	55.4	48.7	43.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	52.1	54.7	59.0	62.6	67.1	69.2	70.4	65.8	62.9	57.6	51.7
		MCDB	60.5	62.3	70.1	74.9	81.6	87.2	89.5	88.6	82.7	74.0	67.5	57.7
	2%	WB	47.4	49.4	51.8	55.7	60.3	64.5	66.9	67.6	63.8	60.3	54.2	48.8
		MCDB	56.6	59.7	66.9	69.8	78.7	82.7	89.3	86.8	80.3	72.6	65.1	55.8
	5%	WB	45.0	47.1	50.1	53.7	58.9	62.4	65.4	65.9	62.4	58.5	52.0	46.9
		MCDB	54.1	58.1	64.4	67.6	77.7	79.8	87.5	84.9	78.6	69.7	63.2	54.0
	10%	WB	42.9	45.0	48.7	51.8	57.4	60.9	63.9	64.3	61.1	56.8	49.9	45.2
		MCDB	52.1	56.5	61.9	65.7	74.3	78.2	85.6	83.2	75.8	67.5	59.6	52.5
Mean Daily Temperature Range	5% DB	MDBR	17.4	18.9	19.9	21.2	21.9	24.3	25.2	24.4	21.8	19.3	18.2	16.9
		MCDBR	21.5	24.0	25.0	26.6	26.1	27.8	27.3	27.1	25.5	22.9	22.1	21.1
		MCWBR	11.8	12.2	11.3	11.5	10.0	8.5	8.9	8.8	9.3	10.2	11.2	11.4
	5% WB	MCDBR	19.7	21.3	22.4	24.6	24.8	25.6	25.8	25.1	23.4	20.2	20.2	17.7
		MCWBR	11.9	12.0	10.8	11.8	10.4	9.7	10.2	10.2	9.8	10.7	11.5	10.9
Clear-Sky Solar Irradiance	taub	0.242	0.274	0.322	0.345	0.382	0.417	0.478	0.446	0.381	0.339	0.280	0.251	
	taud	2.645	2.521	2.382	2.334	2.243	2.132	1.940	2.052	2.271	2.402	2.563	2.619	
	Ebn at Noon	319	315	304	299	287	275	258	265	280	285	299	309	
	Edh at Noon	23	29	36	40	44	49	60	53	40	33	25	23	
All-Sky Solar Radiation	RadAvg	1112	1406	1774	2096	2290	2475	2353	2126	1834	1447	1146	1021	
	RadStd	93	103	103	137	117	107	89	85	106	80	88	68	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.70	N/A	N/A	+2.30	N/A	N/A	N/A	N/A	+351	+344

Nomenclature: See separate page